Spatio-temporal dynamics of high-Andean wetland vegetation vigor: SARIMA modelling and spatial autocorrelation analysis of satellite indices in Carabaya, Puno, Peru (2019–2025)

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Abstract

High-Andean wetlands (bofedales) are high-elevation ecosystems that play a critical role in hydrological regulation and the sustainability of alpaca grazing systems, and face increasing pressure from climate variability and anthropogenic activity. Peru's National Wetland Inventory reports a reduction of up to 30% in wetland area in northern Puno, including Carabaya Province, yet this figure describes area loss and not the spectral condition of the remaining vegetation. This study characterizes the spatio-temporal dynamics of vegetation vigor in the wetlands of Carabaya through two complementary components. The spatial component assesses the dependence structure (semivariogram and kriging) and spatial autocorrelation (global Moran's I and local Getis-Ord Gi*) of mean vegetation vigor (2019–2025) at 300 sampling points. The temporal component fits SARIMA models to monthly time series of NDVI, SAVI, EVI, and NDMI for the wetland and an adjacent grassland control area, benchmarked against ETS and TBATS models through rolling-origin cross-validation (RMSE), and assesses trend, using both conventional and seasonal Mann–Kendall tests, and the relationship with precipitation. Results show significant spatial autocorrelation for all three indices evaluated (Moran's I, $p < 0.001$), with geographic coincidence of hot-spot and cold-spot zones across indices, and NDVI and EVI semivariogram ranges of 9.6 and 12.5 km, respectively. In the temporal component, all SARIMA models produced residuals consistent with white noise (Ljung–Box $p > 0.6$), and no single forecasting framework (SARIMA, ETS, TBATS) consistently outperformed the others across the eight series. Wetland series showed no significant trend under either Mann–Kendall test, whereas control SAVI showed a significant increasing trend under the seasonal test ($\tau = 0.214$, $p = 0.019$), suggesting that vegetation change, if occurring, may be more evident in the non-wetland grassland than in the wetland core.

Keywords: high-Andean wetlands, time series, SARIMA, geostatistics, spatial autocorrelation, Moran's I, Getis-Ord, NDVI, NDMI, Sentinel-2, Puno, Peru

Highlights

- Sentinel-2 indices were integrated with spatial and temporal statistical analysis.
- Significant spatial autocorrelation was detected across wetland vegetation indices.
- SARIMA, ETS, and TBATS showed no universal forecasting superiority.
- Wetland vegetation vigor showed no significant monotonic decline during 2019–2025.
- Wetland extent and vegetation condition provide complementary indicators of ecosystem change.

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1. Introduction

High-Andean wetlands (bofedales) are mountain wetland ecosystems occurring above 3800 m a.s.l. [21, 25], characterized by permanently hydromorphic vegetation that regulates the local hydrological cycle [2], sustains South American camelid pastoralism, and constitutes a biodiversity reservoir [17]. Puno holds the largest wetland extent in Peru according to the National Wetland Inventory, although the same inventory reports a reduction of up to 30% in wetland area in its northern zone — comprising Carabaya, San Antonio de Putina, and Melgar — attributed to climate variability and anthropogenic pressure [17].

This figure, however, is an *area*-based estimate obtained through land-cover classification at a single census period, capturing neither the *spatial distribution* nor the *temporal dynamics* of the vegetation vigor remaining within the ecosystem. A monitoring approach complementary to point-in-time inventories requires, on one hand, characterizing whether vegetation vigor exhibits identifiable spatial structure and clustering, and on the other, evaluating its temporal evolution and the presence of trend.

Remote sensing through spectral indices has been widely used for monitoring wetlands and agricultural vegetation. NDVI [24] is the most widely used index, although it saturates under dense cover and is sensitive to soil background; SAVI [13] corrects for soil effects, and EVI [14] additionally corrects atmospheric effects and saturation. NDMI, derived from the water index proposed by Gao [10], is sensitive to foliar moisture content, making it particularly relevant for water-regime-dependent ecosystems such as bofedales.

Previous regional geostatistical studies have applied spatial interpolation, such as indicator kriging on NDVI [20], while recent work has addressed satellite crop monitoring through supervised machine learning, showing that phenological signature is more informative than the point value of the index [3]. These supervised approaches, however, require extensive ground-truth data not available for Peruvian highland wetlands, motivating an unsupervised statistical approach based on spatial and time-series analysis.

SARIMA models [1, 4] constitute a robust tool for describing series with trend and seasonality without requiring labeled data, implemented through the automatic selection procedure of Hyndman and Khandakar [16]. To assess whether the choice of SARIMA is empirically justified rather than arbitrary, this study benchmarks it against two alternative models widely used in the forecasting literature: exponential smoothing state space models (ETS) [15] and Trigonometric, Box-Cox, ARMA, Trend and Seasonal (TBATS) models [6].

In this context, the present study aims to characterize the spatio-temporal dynamics of vegetation vigor in the wetlands of Carabaya Province (Puno, Peru) by evaluating: (i) the spatial dependence structure and autocorrelation of mean vegetation vigor through semivariogram, kriging, Moran's I, and Getis-Ord Gi*; (ii) the monthly temporal dynamics (2019–2025) of four vegetation indices through SARIMA models, benchmarked against ETS and TBATS, and compared with a control grassland area; and (iii) the presence of significant trend, assessed through both conventional and seasonal Mann–Kendall tests, and its relationship with accumulated precipitation.

2. Study Area

The study was conducted in Carabaya Province, northern sector of the Puno Department, southeastern Peru (Figure 1). Carabaya holds 18,591.7 ha of wetland across 9,920 discrete geomorphological units [17], approximately 7.1% of Puno's total wetland area. The province presents a high-Andean climatic regime with marked seasonality, concentrating precipitation between December and March, with relatively arid conditions between May and August.

3. Materials and Methods

3.1. Data sources

Wetland spatial delineation was obtained from the National Wetland Inventory of Peru, second edition 2023 (INAIGEM),

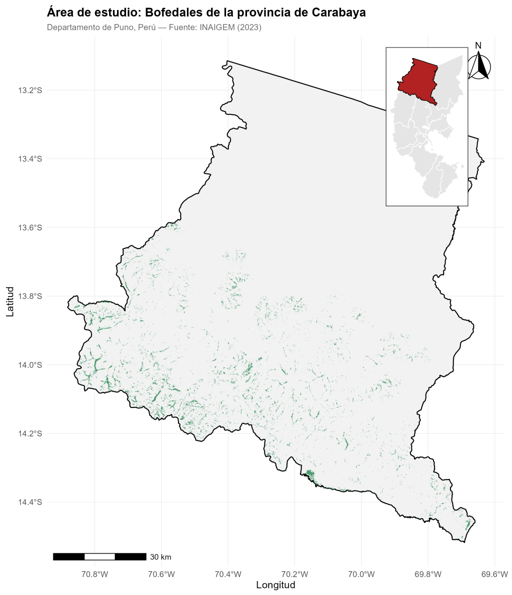

Figure 1: Study area: distribution of wetlands (green) in Carabaya Province, Puno, Peru. The inset in the upper-right corner shows the location of Carabaya (red) within Puno. Source: prepared by the authors based on INAIGEM (2023).

based on classification of 2021 Sentinel-2 imagery [17], spatially clipped to Carabaya via QGIS (9,920 polygons). Vegetation indices were derived from Sentinel-2 Level 2A Surface Reflectance Harmonized data [8] processed in Google Earth Engine, generating monthly median composites filtered to below 40% cloud cover. Reliable coverage begins in January 2019, setting the analysis period between January 2019 and December 2025 (84 months). Monthly accumulated precipitation from the CHIRPS daily product [9] was used as a complementary climatic variable.

3.2. Vegetation index calculation

For each monthly composite the following indices were computed from surface reflectance bands scaled to their physical range (0–1):

$$\text{NDVI} = \frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED}} \tag{1}$$

$$\text{SAVI} = (1 + L)\frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + \rho_{RED} + L}, \quad L = 0.5 \tag{2}$$

$$\text{EVI} = 2.5\frac{\rho_{NIR} - \rho_{RED}}{\rho_{NIR} + 6\rho_{RED} - 7.5\rho_{BLUE} + 1} \tag{3}$$

$$\text{NDMI} = \frac{\rho_{NIR} - \rho_{SWIR}}{\rho_{NIR} + \rho_{SWIR}} \tag{4}$$

where ρ_{BLUE}, ρ_{RED}, ρ_{NIR}, and ρ_{SWIR} correspond to Sentinel-2 bands B2, B4, B8, and B11.

3.3. Spatial sampling design

Given the high fragmentation of the wetland inventory (9,920 discrete polygons), a sample of 300 points was generated at the centroids of a random subset of wetland polygons, and an equivalent set of control points was generated by shifting each centroid 1,000 m northward, intended to represent adjacent non-wetland high-Andean grassland. Because control points were defined through a fixed geometric displacement rather than independent land-cover verification, this design should be regarded as a practical approximation; the possibility that some control points fall on secondary wetland patches, water bodies, or other non-grassland cover was not formally excluded and is addressed as a limitation (Section 5.1). Indices were extracted at each point via median reduction at 10 m resolution.

3.4. Spatial structure and interpolation

To characterize the spatial dependence of mean vegetation vigor (2019–2025) across the 300 wetland points, experimental semivariograms of NDVI and EVI were computed, fitting spherical, exponential, and Gaussian models by least squares and selecting the model with the lowest sum of squared errors (SSErr) for each index. The selected NDVI model was used to generate an interpolated surface via ordinary kriging on a 500 m regular grid, clipped to the wetland area of Carabaya.

3.5. Spatial autocorrelation

Global spatial autocorrelation of mean vegetation vigor was assessed using Moran's I [23], with a distance-based neighborhood matrix using a radius equal to the NDVI semivariogram range (9,630 m), under the assumption that this distance approximates the structural spatial dependence limit of the variable. The local Getis-Ord Gi* statistic [11] was additionally applied to identify significant spatial clusters of high (hot spots) and low (cold spots) values, classified using standard significance thresholds ($z > 1.96$ and $z > 2.58$ for 95% and 99%, respectively).

3.6. Time-series analysis

Missing monthly values due to the absence of valid images were linearly interpolated using neighboring observations, with series endpoints extrapolated using the nearest valid value. This interpolation was performed once on the full series prior to model fitting and cross-validation; the associated methodological implications are discussed in Section 5.1. Each series was decomposed using the STL procedure [5], and stationarity was assessed via the Augmented Dickey–Fuller [7] and KPSS [19] tests, determining the differencing order using the `ndiffs` and `nsdiffs` functions of the R *forecast* package.

3.7. SARIMA models

A SARIMA$(p, d, q)(P, D, Q)_{12}$ model was fitted to each of the eight series (four indices × two land-cover types) via exhaustive search [16], with AIC used for within-framework order selection. Model adequacy was assessed through the Ljung–Box test on residuals.

3.8. Alternative forecasting models

As a robustness check on the choice of SARIMA, exponential smoothing state space models (ETS) [15] and TBATS models [6] were fitted to the same eight series using the `ets()` and `tbats()` functions of the *forecast* package.

3.9. Forecasting model comparison

AIC was used for within-framework model selection (i.e., choosing the SARIMA order), whereas out-of-sample comparison across SARIMA, ETS, and TBATS was based on rolling-origin, one-step-ahead time-series cross-validation, since AIC values are not directly comparable across models with different likelihood formulations. For each series, the model was repeatedly refit using an expanding training window, generating a one-step-ahead ($h = 1$) forecast at each origin t, and the root mean squared error was computed as:

$$\mathrm{RMSE}_{CV} = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(\hat{y}_{t|t-1} - y_t)^2} \qquad (5)$$

where $\hat{y}_{t|t-1}$ denotes the forecast for month t generated using only information available up to $t - 1$, and n is the number of forecast origins evaluated (implemented via the `tsCV()` function of the *forecast* package, with a minimum training window of 24 months).

3.10. Trend analysis and relationship with precipitation

The non-parametric Mann–Kendall test [22, 18] was applied to each series to assess monotonic trend ($\alpha = 0.05$). Because the series exhibit strong seasonality, which can bias the conventional Mann–Kendall test, the Seasonal Mann–Kendall test [12], which evaluates trend within each calendar month separately before combining results, was additionally applied as a robustness check. The relationship with precipitation was assessed via cross-correlation analysis (CCF) on deseasonalized residuals of both series, and as a further robustness analysis SARIMAX models with precipitation lagged 0 to 3 months were fitted, selecting the lowest-AIC lag.

4. Results

4.1. Spatial autocorrelation and hot spots

Global Moran's I was positive and statistically significant for all three indices evaluated (NDVI: $I = 0.077$, $p < 0.001$; EVI: $I = 0.101$, $p < 0.001$; NDMI: $I = 0.138$, $p < 0.001$), confirming that vegetation vigor exhibits spatial autocorrelation: nearby wetland points resemble each other significantly more than distant ones. The relatively low magnitude of the statistic, combined with the high semivariogram nugget, indicates that this spatial structure coexists with considerable short-range heterogeneity. NDMI showed the strongest autocorrelation of the three indices, consistent with its greater sensitivity to local hydrological conditions.

The Getis-Ord Gi* analysis identified significant spatial clusters of high and low vigor for all three indices (Figure 2), with notable geographic coincidence: the south-central sector of the province (approximately 70.3°–70.5°W, 14.0°–14.2°S) was consistently identified as a cold spot across all three indices, while a nearby sector (70.4°–70.5°W, 13.9°–14.0°S) was identified as a hot spot in EVI and NDMI. This convergence across independent indices reinforces the robustness of the identified spatial patterns.

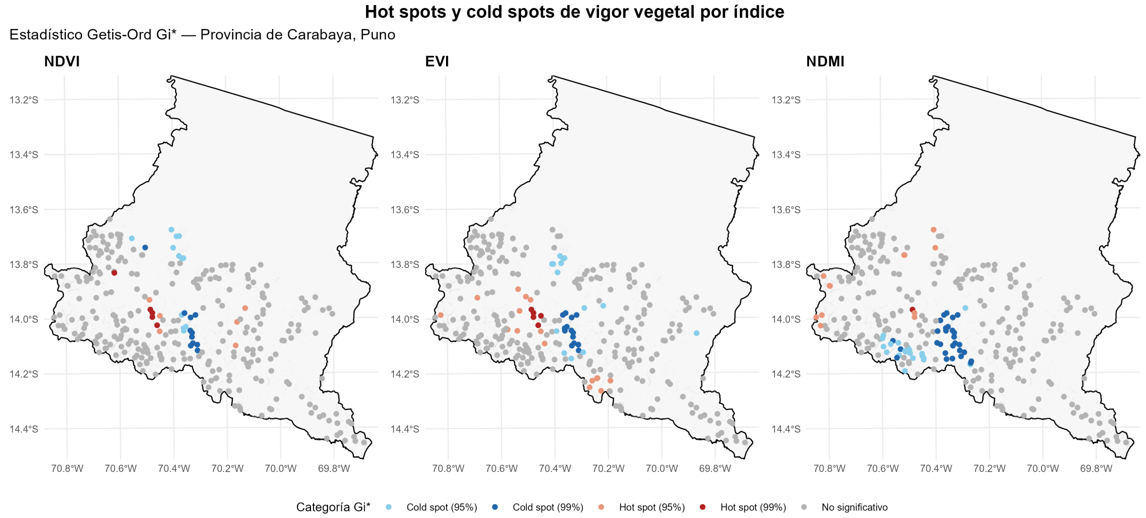

Figure 2: Hot spots and cold spots of vegetation vigor (NDVI, EVI, NDMI) according to the Getis-Ord Gi* statistic, Carabaya Province, Puno.

4.2. Descriptive statistics

Table 1: Descriptive statistics of vegetation indices (2019–2025).

Variable	Mean	SD	Min	Max
Wetland NDVI	0.507	0.112	0.264	0.707
Control NDVI	0.280	0.068	0.150	0.412
Wetland SAVI	0.302	0.059	0.168	0.420
Control SAVI	0.156	0.034	0.084	0.226
Wetland EVI	0.316	0.062	0.189	0.419
Control EVI	0.179	0.057	0.078	0.359
Wetland NDMI	0.073	0.127	−0.146	0.325
Control NDMI	−0.129	0.118	−0.247	0.279

Wetlands showed systematically higher mean values than the control area for NDVI, EVI, and NDMI (Table 1), consistent with greater moisture content and persistent vegetation cover.

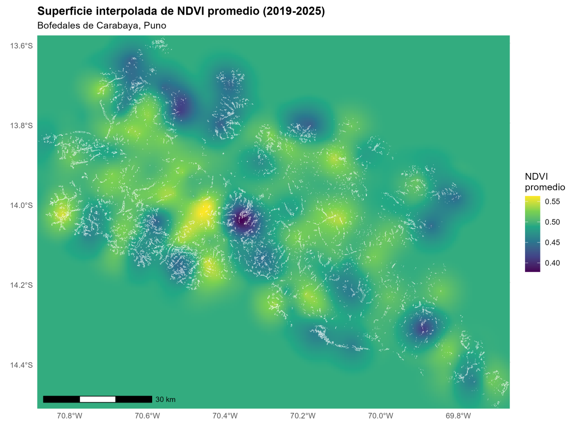

Figure 3: Mean NDVI surface (2019–2025) interpolated via ordinary kriging, Carabaya Province, Puno.

4.3. Spatial structure

The NDVI experimental semivariogram was best fitted by a spherical model (nugget = 0.0049, partial sill = 0.0022, range = $9,630$ m), while the EVI semivariogram was best fitted by a Gaussian model (nugget = 0.0033, partial sill = 0.0006, range = $12,544$ m). Both fitted models showed a well-defined sill, indicating that spatial dependence of vegetation vigor, as captured by these two indices, is lost beyond approximately 9.6–12.5 km. The relatively high nugget observed in both cases indicates substantial short-range variability, which may reflect the fragmented and heterogeneous environmental conditions characteristic of high-Andean wetlands, although the semivariogram alone does not establish the specific cause of this variability. The interpolated NDVI surface via ordinary kriging (Figure 3) shows a gradient of higher vegetation vigor in the west-central sector of the province.

4.4. Seasonality

All eight series showed marked, regular seasonality, with maxima between March and May and minima between August and October (Figure 5), consistent with the high-Andean climatic regime. STL decomposition (Figure 4) revealed a dominant seasonal component and a low-magnitude residual with no apparent pattern, an analogous pattern to the other indices evaluated.

4.5. SARIMA models

All eight fitted models produced residuals consistent with white noise (Ljung–Box $p > 0.60$ in all cases; Table 2). Wetland EVI obtained the lowest cross-validation RMSE (0.0306), followed by wetland SAVI (0.0349), while control NDMI showed the highest error (0.0945).

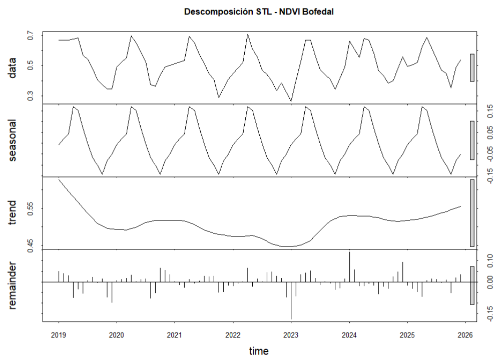

Figure 4: STL decomposition of the NDVI series in the wetland area.

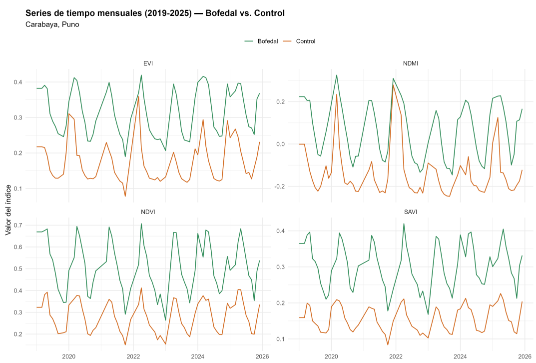

Figure 5: Monthly series (2019–2025) of NDVI, SAVI, EVI, and NDMI in the wetland area (green) and the control area (orange), Carabaya Province, Puno.

Table 2: Fitted SARIMA models and comparative performance.

Series	SARIMA order	AIC	$RMSE_{CV}$
Wetland NDVI	$(1,0,0)(1,1,1)_{12}$	−198.1	0.0730
Wetland SAVI	$(0,0,2)(0,1,1)_{12}$	−289.9	0.0349
Wetland EVI	$(2,0,0)(0,1,1)_{12}$	−318.3	0.0306
Wetland NDMI	$(1,0,1)(2,1,0)_{12}$	−229.5	0.0608
Control NDVI	$(1,0,0)(0,1,1)_{12}$	−290.2	0.0402
Control SAVI	$(1,0,0)(1,1,0)_{12}$	−388.0	0.0200
Control EVI	$(1,0,1)(1,1,0)_{12}$	−303.9	0.0395
Control NDMI	(2,0,1)	−207.6	0.0945

4.6. *Comparison of SARIMA, ETS, and TBATS*

No forecasting framework outperformed the others across the eight series (Table 3): SARIMA achieved the lowest RMSE in three of eight series, ETS in three of eight, and TBATS in two of eight. SARIMA was retained as the modelling framework because it provided competitive predictive performance across the evaluated series, rather than because it systematically outperformed the alternative methods. The higher TBATS error for wetland NDMI (RMSE = 0.139, more than twice the error obtained with SARIMA and ETS) is noteworthy in light of the high nugget observed for this index in the spatial analysis (Section 4.3). This concurrent spatial and temporal variability suggests that NDMI may exhibit greater heterogeneity that is not equally captured by all forecasting formulations, without implying a direct causal relationship between the two analyses.

Table 3: Cross-validation RMSE comparison between SARIMA, ETS, and TBATS.

Series	SARIMA	ETS	TBATS	Best
Wetland NDVI	0.0730	0.0705	0.0651	TBATS
Wetland SAVI	0.0349	0.0426	0.0367	SARIMA
Wetland EVI	0.0306	0.0358	0.0292	TBATS
Wetland NDMI	0.0608	0.0605	0.1393	ETS
Control NDVI	0.0402	0.0421	0.0439	SARIMA
Control SAVI	0.0200	0.0227	0.0238	SARIMA
Control EVI	0.0395	0.0365	0.0371	ETS
Control NDMI	0.0945	0.0796	0.0988	ETS

4.7. *Trend*

Under the conventional Mann–Kendall test, none of the eight series showed a significant monotonic trend ($p > 0.05$ in all cases; Table 4). Under the Seasonal Mann–Kendall test (Table 5), all four wetland series and three of four control series likewise showed no significant trend; the exception was control SAVI, which showed a significant increasing trend ($\tau = 0.214$, $p = 0.019$). This result was not detected by the conventional test, illustrating the value of the seasonal correction, and indicates that if vegetation change is occurring in Carabaya, it may be more evident in the non-wetland grassland than within the wetland areas sampled in this study.

Table 4: Conventional Mann–Kendall trend test.

Series	τ	p	Trend
Wetland NDVI	−0.053	0.477	Not significant
Wetland SAVI	−0.010	0.896	Not significant
Wetland EVI	0.000	1.000	Not significant
Wetland NDMI	−0.098	0.186	Not significant
Control NDVI	−0.004	0.957	Not significant
Control SAVI	0.031	0.677	Not significant
Control EVI	−0.020	0.787	Not significant
Control NDMI	−0.123	0.098	Not significant

Table 5: Seasonal Mann–Kendall trend test.

Series	τ	p	Trend
Wetland NDVI	−0.008	0.931	Not significant
Wetland SAVI	0.095	0.298	Not significant
Wetland EVI	0.135	0.141	Not significant
Wetland NDMI	−0.167	0.069	Not significant
Control NDVI	0.143	0.119	Not significant
Control SAVI	0.214	0.019	Increasing (significant)
Control EVI	0.071	0.435	Not significant
Control NDMI	−0.119	0.193	Not significant

4.8. *Relationship with precipitation*

Cross-correlation analysis on deseasonalized residuals showed no consistent dominant lag (Figure 6). SARIMAX models did not yield significant precipitation coefficients for any of the three indices evaluated, indicating that its effect is already captured by the seasonal component.

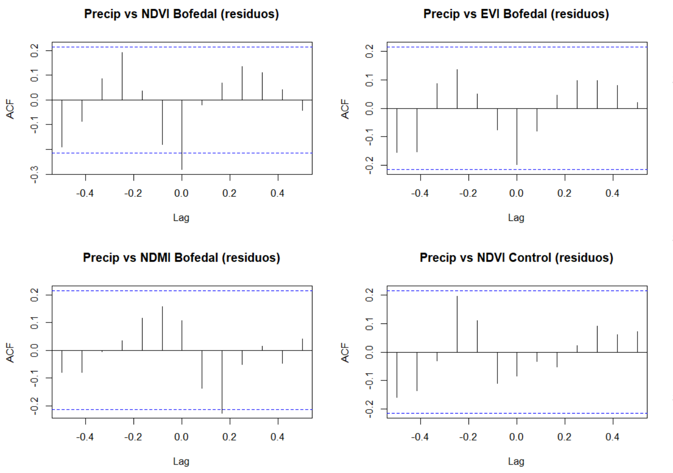

Figure 6: Cross-correlation between deseasonalized residuals of precipitation and of NDVI, EVI, and NDMI (wetland) and NDVI (control).

4.9. *Forecasting*

The 24-month forecasts (2026–2027) project the continuity of the observed seasonal pattern, with no evidence of structural change (Figure 7).

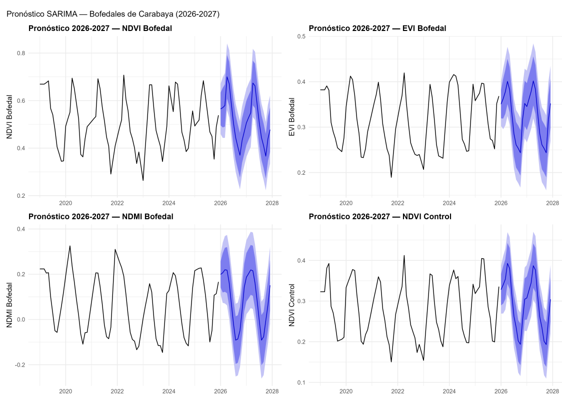

Figure 7: 24-month SARIMA forecast (2026–2027) for NDVI, EVI, and NDMI in the wetland, and NDVI in the control area.

5. Discussion

The spatial and temporal results of this study characterize the condition of the Carabaya wetlands from two complementary dimensions. Spatially, a real dependence structure exists for NDVI and EVI (semivariogram ranges of 9.6 and 12.5 km, respectively) along with significant autocorrelation across all three indices evaluated (Moran's I), with identifiable and cross-index coincident clusters of high and low vigor (Getis-Ord Gi*) — evidence that vegetation vigor is not randomly distributed but responds to local conditions shared among nearby wetland patches. Temporally, wetland vegetation vigor shows marked seasonality and the absence of significant trend within the 2019–2025 window, under both conventional and seasonal Mann–Kendall tests.

This finding should be interpreted as complementary, not contradictory, to the 30% wetland area reduction reported at the inventory level for northern Puno [17]. Both metrics describe distinct processes: total ecosystem extent over a multi-year horizon, versus the spectral vegetation condition of the remaining cover within a six-year window. It is plausible that area loss concentrates at margins of lower relative moisture, while the remaining wetland areas represented by the sampled polygons in this study maintain hydrological conditions sufficient to sustain stable vegetation vigor, particularly in the sectors identified as hot spots in the spatial analysis. The sectors identified as cold spots, in contrast, could constitute priority zones for finer-grained temporal monitoring or management intervention. The significant increasing trend detected in control SAVI under the seasonal test further suggests that, within the analyzed window, detectable vegetation change in Carabaya may be more associated with the surrounding grassland matrix than with the wetland core.

The comparison between SARIMA, ETS, and TBATS did not show a consistently superior framework across the eight series evaluated, supporting model selection according to the characteristics of each index and land-cover class rather than the a priori adoption of a single method. The predictive superiority of EVI and SAVI over NDVI is consistent with their methodological design, oriented toward correcting saturation and soil effects [13, 14]. The greater discriminant capacity of NDMI between wetland and control is consistent with its sensitivity to water content [10], the functionally defining variable of the wetland, and its greater spatial autocorrelation (Section 4.1) suggests that the ecosystem's hydrological variability has a joint spatial and temporal expression.

The absence of a significant relationship between monthly precipitation and the indices, once shared seasonality is removed, suggests that the wetland's hydrological dynamics operate with a buffering capacity that exceeds the monthly scale, consistent with its function of gradual water storage and release.

5.1. *Limitations*

This study relied on the official INAIGEM (2023) delineation without independent field validation. The control area was defined through a fixed geometric displacement rather than field or auxiliary land-cover verification, and the possibility that some control points do not fully correspond to non-wetland grassland was not formally excluded; this represents a methodological approximation to be addressed through field validation in future work. Missing values were interpolated once across the full series prior to model fitting and cross-validation; because a small number of these interpolated points could, in principle, have used information from both preceding and following months, this procedure may introduce minor optimistic bias in the reported cross-validation errors, particularly for the isolated gaps outside the 2017–2018 period excluded from the analysis. Given the limited number and non-consecutive nature of these gaps (17 of 84 months), this effect is expected to be small, but it is acknowledged as a limitation rather than fully eliminated. The neighborhood radius used for Moran's I and Getis-Ord Gi* was based on the NDVI semivariogram range; while statistically grounded, other neighborhood criteria could yield slightly different results. The temporal window (2019–2025) is constrained by the availability of reliable Sentinel-2 Level 2A coverage.

6. Conclusions

Vegetation vigor in the Carabaya wetlands exhibits significant spatial structure and autocorrelation, with cross-index coincident clusters of high and low vigor, and showed no signs of progressive temporal degradation between 2019 and 2025 under either conventional or seasonal Mann–Kendall tests, in contrast with the area loss reported at the inventory level. The comparison among SARIMA, ETS, and TBATS showed that no single forecasting framework consistently outperformed the others across all series, supporting model selection according to the characteristics of each index and land-cover class rather than the assumption that any one method is universally best. The combined use of EVI and SAVI is recommended for routine vegetation vigor monitoring, and NDMI for tracking the wetland's hydrological function, particularly in sectors identified as cold spots through the Getis-Ord Gi* analysis, which constitute priority candidates for higher-temporal-resolution monitoring and field validation. Future studies should extend the temporal window as the Sentinel-2 historical archive consolidates, incorporate independent field validation of control areas, and develop explicit spatio-temporal models (e.g., space-time kriging) that formally integrate both dimensions analyzed separately in this study.

CRediT authorship contribution statement

Denilzon Robinho Tapara Tristan: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing.

Data Availability Statement

Vegetation index and climate variable data were extracted through Google Earth Engine from the public Sentinel-2 SR Harmonized and CHIRPS Daily collections. The wetland delineation layer corresponds to Peru's National Wetland Inventory (INAIGEM, 2023), which is publicly available.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with English language translation, language refinement, and organization of the written presentation. The author reviewed and edited the AI-assisted content and takes full responsibility for the accuracy, integrity, and final content of the manuscript.

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